Metronautica Tibble

A sample-level acceleration data contract for retaining signals and supporting later analyses

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1 Required data contract

The Metronautica Tibble retains sample timestamps and triaxial acceleration for later analysis. Its four data columns are the only required fields. Metadata, diagnostics, reports and bundle wrappers described below are optional implementations of the Metronautica pipeline.

1.1 Structure and representation

`as_accel_tbl()` constructs an `accel_tbl` inheriting from `tbl_df`, with an empty meta list by default. Compatible four-column tibbles and data frames also conform; no class tag, meta attribute or version field is required.

Required column names and order: time, `ax_g`, `ay_g`, `az_g`. Columns have equal length, with one row per decoded observation.

Axes are numeric acceleration in g; 1 g = 9.80665 m/s². time is `POSIXct`. UTC is the storage convention; the constructor preserves supplied instants, timezone and sample order. Invalid observations remain available for diagnostics. Structural conformance does not establish signal quality, calibration or suitability for an analysis.

Minimal construction without metadata

```
x <- as_accel_tbl(time, ax_g, ay_g, az_g)
```

1.2 Required data columns

Field	Type	Description
`time`	POSIXct	Sample timestamp. Parsers store UTC. Duplicates, backward steps and invalid timestamps are retained for diagnostics; downstream functions require suitable timing.
`ax_g`	numeric	Acceleration on device X axis in g.
`ay_g`	numeric	Acceleration on device Y axis in g.
`az_g`	numeric	Acceleration on device Z axis in g.

2 Optional pipeline metadata

Optional metadata is stored as attr(x, 'meta'), a list with file, acquisition and processing sections read by `accel_meta(x)`. Every field below is a pipeline implementation field and may be absent when unknown or not produced. Parser bundles use list(data = x, meta = `accel_meta(x)`), with synchronised metadata. Some analyses require settings or explicit arguments beyond the core contract.

2.1 File metadata

Field	Type	Description
source_path	character	Input file path used for parsing.
source_name	character	Basename of the input file.
format	character	Detected or declared native format, e.g. actigraph_leap_agsd.
bytes	numeric	File size in bytes.
hash_sha256	character	SHA-256 digest of the input file for integrity and reproducibility.
parser	character	Primary parsing engine (package or spec label).
parser_version	character	Package version or spec version string for traceability.
parsed_utc	character	UTC timestamp when the file was parsed.
warnings	character vector	Non-fatal parsing warnings (empty if none).
auto_parser	character	Automatic parser dispatcher name.
auto_parser_version	character	Dispatcher version; device-specific decoding details may also be recorded under processing.

2.2 Acquisition metadata

Field	Type	Description
start_utc	POSIXct (UTC)	Minimum timestamp in decoded data.
end_utc	POSIXct (UTC)	Maximum timestamp in decoded data.
fs_nominal_hz	numeric	Nominal sampling rate from device header if available.
fs_derived_hz	numeric	Usually inverse median positive inter-sample interval. LEAP AGSD uses its nominal packet-reconstruction rate. Neither independently validates the device clock.
full_scale_range_g	numeric	Full-scale range in g (e.g., +/-8 g becomes 16 g). User-supplied or read from header when supported.
units	character	Acceleration unit label, fixed to 'g'.
device_vendor	character	Device vendor name (e.g., ActivInsights, SENS, Axivity, ActiGraph).
device_model	character	Device model family (e.g., GENEActiv, SENS motion, AX3/AX6, GT3X/GT3X+).
device_id	character	Device serial or unique identifier when available.

Field	Type	Description
sensor_vendor	character	Optional sensor vendor field (user-supplied).
sensor_model	character	Optional sensor model field (user-supplied).
time_tz_declared	character	Time-zone declaration retained from the source, if available.
time_tz_parsed	character	Time-zone interpretation used during parsing; canonical output is UTC.
time_tz_source	character	Evidence for timestamp interpretation, e.g. container_epoch or unknown.
time_shift_s	numeric	Explicit shift applied to decoded timestamps, in seconds.

2.3 Processing metadata

Field	Type	Description
parser_library	list	Parser/dispatcher name and version.
calibration	list	Decoding status, scale/calibration application, coefficient sources, applied_by, details, notes and caveats. Application does not establish physical traceability.
decode_notes	character	Critical decoding decisions and current QC scope.
qc	list	Post-decode primary-five signal diagnostics (Section 3).
history	list	Processing events, where recorded; e.g. clipping, filtering or researcher calibration.
reports	list	Attached static/dynamic assessment lists when enabled (Section 4).

3 Optional signal diagnostics

Diagnostics are the recommended minimum workflow assessment, not required contract fields. The primary-five ruleset describes integrity, sample timing, axis-zero proportions, unusual cross-axis sequences and near-limit observations. `parse_raw_auto` attaches it at `meta$processing$qc` by default (qc = TRUE); `as_accel_tbl` does not. Use `attach_qc` for an existing table. Diagnostics do not remove, repair or certify samples; unknown range makes near-limit results unavailable.

Field	Type	Description
`ruleset`	character	Diagnostic definition identifier: primary-five.
`n_samples`	integer	Number of observations in the assessed table.
`missing_time`	numeric	Number of non-finite sample timestamps.
`duplicate_time`	numeric	Duplicated finite timestamps; each repeat after its first occurrence is counted.
`backward_time`	numeric	Consecutive finite time steps with dt < 0; duplicates are reported separately.
`invalid_axis`	numeric	Total non-finite X/Y/Z values.
`median_dt_ms, iqr_dt_ms, longest_dt_ms`	numeric	Median, IQR and maximum finite positive inter-sample interval, in milliseconds.
`zero_x_n, zero_y_n, zero_z_n`	numeric	Counts of finite exact zeros on each axis.
`finite_x_n, finite_y_n, finite_z_n`	numeric	Per-axis finite-value denominators.
`zero_x_pct, zero_y_pct, zero_z_pct`	numeric	100 times exact-zero count divided by that axis's finite-value count.
`all_zero_samples, all_zero_sequence`	numeric, logical	Count of simultaneous finite X=Y=Z=0 samples and whether a qualifying adjacent pair exists (Section 3.1).
`equal_nonzero_samples, equal_nonzero_sequence`	numeric, logical	Count of finite X=Y=Z≠0 samples and whether a qualifying adjacent pair exists.
`limit_g, near_limit_fraction`	numeric	Per-axis limit = full_scale_range_g/2; near-limit fraction = 0.98. Invalid/unknown range gives NA limit.
`near_limit_n, finite_axis_n`	numeric	Pooled count of finite axis values with |axis| ≥ 0.98 × limit_g, and total finite X/Y/Z values.
`near_limit_pct`	numeric	100 × near_limit_n/finite_axis_n; NA when range is unknown or no finite axes exist.

3.1 Sequence checks and scope

A sequence requires at least two adjacent qualifying samples with finite timestamps, finite axes and a positive intervening time step. No minimum-duration run, gap-threshold count or saturation-duration summary is returned. Near-limit observations do not establish clipping or its cause. External subsetting can leave stale diagnostics; package static/dynamic clipping recomputes QC, while filtering/calibration invalidates it.

4 Optional assessment records

Static and dynamic records are optional pipeline attachments. `static_metrology_from_tbl`, `static_noise_summary_from_tbl` and `dynamic_metrology_report` default to `attach_report` = TRUE; FALSE adds no record. Assessments append to `meta$processing$reports$static` or $dynamic, each a list of records. Capture the returned data: inputs are not changed in place. Assessment `schema_version` = "1" versions the report structure independently of this 0.1.0 data contract.

Field	Type	Description
`schema_version, type, function_name`	character	Record schema version 1; static or dynamic; generating function.
`created_utc, software`	POSIXct, list	Assessment creation time in UTC; package and R versions.
`status`	character	current for the assessed source state and selection; historical after package signal/selection changes.
`source, selection`	list	File/acquisition/processing provenance and assessed window/stage bounds and sample counts.
`settings, results, diagnostics`	list	Calculation settings, numeric result tables and assessed-selection QC. Full static reports retain summary, Allan and one_hz tables; noise-only reports retain summary; dynamic reports retain stages and overall.
`reference`	list or NULL	Dynamic reference provenance and assessed selections; NULL for static reports.
`invalidated_by`	character vector	Present after invalidation: selection_changed or signal_or_selection_changed. Records remain as provenance.

Static results return the complete input table as data and the assessed window as `clipped_data`; both carry the record. Dynamic results return data (the full test clip, otherwise its data table or concatenated matched stages) and the enriched `test_clips` bundle. Bundle metadata remain synchronised. Numerical calculations are unchanged by attachment.

Records exclude plots, raw signal copies and recursive parent metadata. Plots remain separate function outputs. Package clipping marks inherited records historical; filtering/calibration does likewise. External edits may leave stale records. A record's scope is its explicit assessed selection, not every sample in the retained signal.

5 Parser implementation notes

GENEActiv (.bin): parsed with GENEAread. `full_scale_range_g` may be inferred from `Measurement_Range` if present, or overridden by user.

SENS (.bin): fixed-record binary with `scale_g_per_lsb` (default 0.0078125 g per LSB). No header; `full_scale_range_g` is user-supplied when required.

Axivity (.cwa): built-in fast decoder for supported layouts, with GGIRread fallback. AX3 CSV exports are also supported.

ActiGraph (.gt3x): parsed with read.gt3x; native precision is the default, with an explicit legacy three-decimal mode. Supply measurement range when absent and needed for diagnostics.

ActiGraph LEAP: direct AGSD decoding applies stored coefficients and retains nominal packet-relative timestamps and gaps; validated layouts are documented in the package. CSV exports are also supported. HBK G-Link CSV is supported for reference recordings.

6 Related documentation

The help topics help("metronautica-tibble") and help("common-data-format") describe this contract.